Abstract Algebra | Quiz 6 | October 14, 2026

Name: _____ | Hammack | Score: _____

1. Find the order of $(8, 8, 8)$ in $\mathbb{Z}_{10} \times \mathbb{Z}_{24} \times \mathbb{Z}_{80}$.

2. Prove or disprove: $U(10) \cong U(12)$.

3. Prove or disprove: $\mathbb{Z} \cong \mathbb{Q}$ (The operation in both groups is $+$.)

4. Prove or disprove: If G and H are groups, then $G \times H \cong H \times G$.